2027 MOU reallocation scenarios, 15 countries (no model changes)

Generated by `audit/reallocation/run_reallocation_2027.py --year 2027` from `allocations_2027_v2.csv`. Preset: America First MOUs, dashboard defaults (no government response, sudden path, death-rate trends on, 400 draws, seed 7). **US bilateral aid only:** the preset's US exits from WHO, Gavi, UNFPA and 23% of the Global Fund are switched off in memory, so the only cut is each country's MOU schedule on US direct channels (bilateral agencies, NGOs); every other donor and every multilateral channel is unchanged. Only the 2027 US MOU amount is replaced, in memory, for the 15 countries in the file; every other year and every other country keeps the schedule in `model_data/country_inputs.csv`. Totals and comparisons cover these 15 countries only.

Step 1. Data check

- `mou_deflate` is off in `model_params.csv` (amounts used nominal, as written).
- Original MOU column vs the model's `mou_us_2027`: 0 of 15 countries differ by more than 0.5% (differences of US$1 or less are rounding). Largest absolute difference: US$0.
- Column totals (US$): Original MOU 2,469,904,824.00; Low Ceiling 2,469,904,823.99; Base Ceiling 2,469,904,824.00; Higher Ceiling 2,469,904,823.99. Spread between columns: US$0.01 (within US$20).

iso3	Country	Sheet Original MOU	Model mou_us_2027	Diff US$	Diff %	mou_us_ref
BWA	Botswana	37,832,582	37,832,582	0	0.000	43,742,433
BDI	Burundi	33,112,000	33,112,000	0	0.000	53,633,354
CMR	Cameroon	91,500,000	91,500,000	0	0.000	120,444,835
SWZ	Eswatini	47,000,000	47,000,000	0	0.000	62,583,783
ETH	Ethiopia	247,000,000	247,000,000	0	0.000	266,100,350
KEN	Kenya	388,294,000	388,294,000	0	0.000	406,525,908
LSO	Lesotho	55,000,000	55,000,000	0	0.000	87,734,264
LBR	Liberia	33,785,480	33,785,480	0	0.000	57,140,755
MDG	Madagascar	32,187,000	32,187,000	0	0.000	79,695,135
MWI	Malawi	165,901,540	165,901,540	0	0.000	239,128,572
MOZ	Mozambique	403,734,000	403,734,000	0	0.000	472,915,607
NGA	Nigeria	455,590,740	455,590,740	0	0.000	511,728,925
RWA	Rwanda	64,075,482	64,075,482	0	0.000	118,139,491
SLE	Sierra Leone	32,700,000	32,700,000	0	0.000	47,288,158
UGA	Uganda	382,192,000	382,192,000	0	0.000	435,160,154

Step 2. Headroom check

The preset's cut for a year is clip(1 - amount / `mou_us_ref`, 0, 1) (`scenarios.py` mou_schedule), so an amount above `mou_us_ref` is a 0% cut and the money above it buys nothing in the model (aid rises count as no loss, audit I-15). The table compares each ceiling's amount with that reference, and for information with the model's 2021-23 US baseline from `model_data/dah_lines.csv` (US direct channels, the rows the cut is applied to; and all US channels). The MOU amounts are nominal US totals from the MOU sheet and the model applies only the ratio, so only the `mou_us_ref` comparison decides whether money is lost to the clip.

iso3	Country	mou_us_ref	US direct 2021-23 (model)	US all channels 2021-23 (model)	Low Ceiling 2027	Base Ceiling 2027	Higher Ceiling 2027
BWA	Botswana	43,742,433	39,403,000	42,625,667	37,832,582	37,832,582	37,832,582
BDI	Burundi	53,633,354	18,131,000	36,964,667	34,638,754	34,638,754	34,638,754
CMR	Cameroon	120,444,835	68,536,333	109,006,000	88,202,316	84,904,631	81,606,947
SWZ	Eswatini	62,583,783	43,912,000	51,681,333	47,000,000	47,000,000	47,000,000
ETH	Ethiopia	266,100,350	104,916,333	180,462,000	247,000,000	247,000,000	247,000,000
KEN	Kenya	406,525,908	262,493,333	323,832,667	368,879,300	349,464,600	330,049,900

iso3	Country	mou_us_ref	US direct 2021-23 (model)	US all channels 2021-23 (model)	Low Ceiling 2027	Base Ceiling 2027	Higher Ceiling 2027
LSO	Lesotho	87,734,264	55,832,000	63,123,667	55,000,000	57,775,084	57,775,084
LBR	Liberia	57,140,755	18,757,333	29,447,667	33,785,480	33,785,480	33,785,480
MDG	Madagascar	79,695,135	18,203,667	37,254,333	32,187,000	32,187,000	32,187,000
MWI	Malawi	239,128,572	154,831,333	219,614,000	165,901,540	165,901,540	165,901,540
MOZ	Mozambique	472,915,607	266,907,667	365,381,000	446,754,039	481,043,759 **(+8,128,152 above ref)**	481,043,759 **(+8,128,152 above ref)**
NGA	Nigeria	511,728,925	269,010,667	403,669,333	433,756,331	411,921,923	390,087,514
RWA	Rwanda	118,139,491	54,217,000	87,496,000	64,075,482	64,075,482	64,075,482
SLE	Sierra Leone	47,288,158	15,085,667	28,816,000	32,700,000	36,327,243	36,327,243
UGA	Uganda	435,160,154	264,462,667	358,174,333	382,192,000	386,046,745	430,593,538

Flags (amount above `mou_us_ref`, money that would count as no gain): Base Ceiling: Mozambique US$8,128,152; Higher Ceiling: Mozambique US$8,128,152 Total affected: US$16,256,304.

For information: 13 of the 15 Original MOU amounts already exceed the model's 2021-23 US direct-channel baseline, because `mou_us_ref` (team sheet, 2021-25 average, all US funding) is larger than the model's IHME-based US direct rows. The model never applies the dollar amounts, only cut fractions, so this is not a gain case.

Step 3. Run

- Original MOU reproduces the model's own MOU schedule for all 15 countries (cuts identical). Deaths 2026-2030 for the 15 countries are 189,629 against 356,197 in the precomputed All Countries file, which includes the US multilateral exits that are switched off here.
- 2027 cut fractions by scenario (1 - amount / mou_us_ref):

iso3	Country	Original MOU	Low Ceiling	Base Ceiling	Higher Ceiling
BWA	Botswana	13.5%	13.5%	13.5%	13.5%
BDI	Burundi	38.3%	35.4%	35.4%	35.4%
CMR	Cameroon	24.0%	26.8%	29.5%	32.2%
SWZ	Eswatini	24.9%	24.9%	24.9%	24.9%
ETH	Ethiopia	7.2%	7.2%	7.2%	7.2%
KEN	Kenya	4.5%	9.3%	14.0%	18.8%
LSO	Lesotho	37.3%	37.3%	34.1%	34.1%
LBR	Liberia	40.9%	40.9%	40.9%	40.9%
MDG	Madagascar	59.6%	59.6%	59.6%	59.6%
MWI	Malawi	30.6%	30.6%	30.6%	30.6%
MOZ	Mozambique	14.6%	5.5%	0.0%	0.0%
NGA	Nigeria	11.0%	15.2%	19.5%	23.8%
RWA	Rwanda	45.8%	45.8%	45.8%	45.8%
SLE	Sierra Leone	30.8%	30.8%	23.2%	23.2%
UGA	Uganda	12.2%	12.2%	11.3%	1.0%

Step 4. Results

4a. Totals, 15 countries

People losing a service are the model's full-cost equivalents in the year (after the 15% continuity assumption), summed over ART, TB treatment, bednets, spraying, malaria treatment and vaccines. Deaths are the central run; ranges are the 2.5th-97.5th percentiles of the draws, and the ranges of changes are paired (same draws in every scenario).

Scenario	Aid lost 2027 US$M	Aid lost 2026-30 US$M	People losing services 2027	Deaths 2026-30 (range)	Deaths in 2027	Further deaths 2031-35	HIV infections 2026-30
Original MOU	275.9	2,358.3	6,914,369	189,629 (135,102-265,790)	17,918	190,857	527,414
Low Ceiling	277.0	2,359.4	7,468,925	189,184 (134,706-264,953)	17,744	190,657	524,174
Base Ceiling	282.9	2,365.3	7,991,831	188,496 (134,163-263,537)	17,743	190,373	520,240
Higher Ceiling	281.7	2,364.1	8,361,756	187,763 (133,778-261,757)	18,152	189,844	516,668

4b. Change vs Original MOU, totals

Scenario	Aid lost 2027 US$M	Aid lost 2026-30 US$M	People losing services 2027 (range)	Deaths 2026-30 (range)	Further deaths 2031-35	HIV infections 2026-30	Deaths HIV	Deaths TB	Deaths Malaria	Deaths Immunization
Low Ceiling	+1.09	+1.09	+554,556 (+327,378 to +601,963)	-446 (-1,062 to -318)	-201	-3,240	-432	-374	+338	+22
Base Ceiling	+6.95	+6.95	+1,077,462 (+650,175 to +1,170,539)	-1,134 (-2,730 to -843)	-485	-7,174	-1,402	-394	+622	+41
Higher Ceiling	+5.77	+5.77	+1,447,387 (+890,408 to +1,581,124)	-1,866 (-4,437 to -1,150)	-1,013	-10,746	-2,845	+42	+849	+88

- **Low Ceiling:** the reallocation is not zero-sum in outcomes: 446 fewer deaths over 2026-2030 (-0.23% of the Original MOU total), 554,556 more people losing a service in 2027, and +1.09 US$M of modelled aid lost in 2027 (the same nominal total of MOU money buys a different amount of modelled aid because each country's cut fraction is applied to its own 2021-23 US direct-channel baseline).
- **Base Ceiling:** the reallocation is not zero-sum in outcomes: 1,134 fewer deaths over 2026-2030 (-0.60% of the Original MOU total), 1,077,462 more people losing a service in 2027, and +6.95 US$M of modelled aid lost in 2027 (the same nominal total of MOU money buys a different amount of modelled aid because each country's cut fraction is applied to its own 2021-23 US direct-channel baseline).
- **Higher Ceiling:** the reallocation is not zero-sum in outcomes: 1,866 fewer deaths over 2026-2030 (-0.98% of the Original MOU total), 1,447,387 more people losing a service in 2027, and +5.77 US$M of modelled aid lost in 2027 (the same nominal total of MOU money buys a different amount of modelled aid because each country's cut fraction is applied to its own 2021-23 US direct-channel baseline).

4c. People losing each service in 2027 and coverage change (totals)

Service	Original MOU people	Original MOU coverage drop pp	Low Ceiling Δ people	Low Ceiling Δ pp	Base Ceiling Δ people	Base Ceiling Δ pp	Higher Ceiling Δ people	Higher Ceiling Δ pp
ART	455,448	4.09	-690	-0.006	+9,287	+0.083	+18,970	+0.170
TB treatment	7,752	0.62	-678	-0.054	-599	-0.048	+446	+0.036
Bednets	5,595,288	0.94	+536,777	+0.091	+1,033,463	+0.174	+1,428,931	+0.241
Spraying	464,390	0.08	-2,718	-0.000	-6,018	-0.001	-41,542	-0.007
Malaria treatment	382,000	0.27	+21,681	+0.016	+40,960	+0.029	+38,913	+0.028
Vaccines	9,491	0.05	+184	+0.001	+369	+0.002	+1,670	+0.008
PMTCT	9,440	2.45	-659	-0.171	-975	-0.253	-2,206	-0.572
HIV prevention (infections averted)	4,529	n/a	-615	n/a	-1,030	n/a	-1,348	n/a

4d. Deaths by disease 2026-2030, totals (range), and the part caused by the 2027 change

Disease	Original MOU deaths 2026-30 (range)	Low Ceiling Δ (paired range)	Base Ceiling Δ (paired range)	Higher Ceiling Δ (paired range)
HIV	118,793 (78,587-201,179)	-432 (-953 to -326)	-1,402 (-2,686 to -1,045)	-2,845 (-4,992 to -1,971)
TB	33,803 (20,050-45,231)	-374 (-600 to -218)	-394 (-721 to -208)	+42 (-293 to +204)
Malaria	34,738 (18,948-39,878)	+338 (+178 to +365)	+622 (+328 to +670)	+849 (+469 to +927)

Disease	Original MOU deaths 2026-30 (range)	Low Ceiling Δ (paired range)	Base Ceiling Δ (paired range)	Higher Ceiling Δ (paired range)
Immunization	2,296 (1,689-4,633)	+22 (+16 to +41)	+41 (+30 to +73)	+88 (+65 to +162)
All	189,629 (135,102-265,790)	-446 (-1,062 to -318)	-1,134 (-2,730 to -843)	-1,866 (-4,437 to -1,150)

Deaths by calendar year, change vs Original MOU (central): the 2027 change also moves deaths in later years (ART cohorts age, TB cases spread 1-2 years later, missed vaccine cohorts die over following years).

Scenario	2026	2027	2028	2029	2030	2031-35
Low Ceiling	+0	-174	-87	-102	-83	-201
Base Ceiling	+0	-174	-313	-342	-304	-485
Higher Ceiling	+0	+234	-678	-722	-701	-1,013

4e. By country: 2027 money change, change in deaths 2026-30 and in people losing services in 2027

Nominal US$ moved is the sheet's change in the country's 2027 MOU amount; modelled aid is the change in the model's aid lost in 2027 (cut fraction x the country's US direct-channel baseline). The per-US$1M columns divide the change in deaths by the money moved, so they are positive in both directions: lives saved per US$1M where money arrives, lives lost per US$1M where it leaves.

Low Ceiling minus Original MOU

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BWA	Botswana	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
BDI	Burundi	+1.53	+0.52	-30 (-52 to -22)	-11	-9,148	-71	+19.5	+57.5
CMR	Cameroon	-3.30	-1.88	+249 (+160 to +412)	+116	+74,552	+626	+75.5	+132.6
SWZ	Eswatini	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
ETH	Ethiopia	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
KEN	Kenya	-19.41	-12.54	+1,114 (+710 to +1,842)	+340	+206,083	+1,689	+57.4	+88.8
LSO	Lesotho	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
LBR	Liberia	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MDG	Madagascar	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MWI	Malawi	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MOZ	Mozambique	+43.02	+24.28	-3,828 (-6,090 to -2,643)	-1,276	-248,304	-9,140	+89.0	+157.7
NGA	Nigeria	-21.83	-11.48	+2,049 (+1,369 to +2,842)	+631	+531,373	+3,656	+93.9	+178.5
RWA	Rwanda	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
SLE	Sierra Leone	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
UGA	Uganda	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
TOTAL	All 15	0.00	-1.09	-446 (-1,062 to -318)	-201	+554,556	-3,240		

Base Ceiling minus Original MOU

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BWA	Botswana	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BDI	Burundi	+1.53	+0.52	-30 (-52 to -22)	-11	-9,148	-71	+19.5	+57.5
CMR	Cameroon	-6.60	-3.75	+452 (+293 to +742)	+207	+149,103	+1,163	+68.5	+120.4
SWZ	Eswatini	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
ETH	Ethiopia	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
KEN	Kenya	-38.83	-25.07	+2,227 (+1,420 to +3,684)	+679	+412,166	+3,379	+57.4	+88.8
LSO	Lesotho	+2.78	+1.77	-205 (-374 to -146)	-131	-3,641	-1,454	+73.7	+115.9
LBR	Liberia	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MDG	Madagascar	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MWI	Malawi	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MOZ	Mozambique	+77.31	+39.05	-7,321 (-11,976 to -5,018)	-2,384	-399,303	-16,744	+94.7	+187.5
NGA	Nigeria	-43.67	-22.96	+4,099 (+2,738 to +5,684)	+1,261	+1,062,747	+7,311	+93.9	+178.6
RWA	Rwanda	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
SLE	Sierra Leone	+3.63	+1.16	-150 (-190 to -101)	-28	-96,206	-187	+41.4	+129.9
UGA	Uganda	+3.85	+2.34	-207 (-332 to -130)	-78	-38,255	-572	+53.6	+88.3
TOTAL	All 15	0.00	-6.95	-1,134 (-2,730 to -843)	-485	+1,077,462	-7,174		

Higher Ceiling minus Original MOU

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BWA	Botswana	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
BDI	Burundi	+1.53	+0.52	-30 (-52 to -22)	-11	-9,148	-71	+19.5	+57.5
CMR	Cameroon	-9.89	-5.63	+549 (+362 to +884)	+239	+223,655	+1,493	+55.5	+97.6
SWZ	Eswatini	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
ETH	Ethiopia	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
KEN	Kenya	-58.24	-37.61	+2,972 (+1,902 to +4,863)	+852	+618,248	+4,666	+51.0	+79.0
LSO	Lesotho	+2.78	+1.77	-205 (-374 to -146)	-131	-3,641	-1,454	+73.7	+115.9
LBR	Liberia	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MDG	Madagascar	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MWI	Malawi	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
MOZ	Mozambique	+77.31	+39.05	-7,321 (-11,976 to -5,018)	-2,384	-399,303	-16,744	+94.7	+187.5
NGA	Nigeria	-65.50	-34.43	+5,679 (+3,844 to +7,717)	+1,636	+1,594,120	+10,012	+86.7	+164.9
RWA	Rwanda	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
SLE	Sierra Leone	+3.63	+1.16	-150 (-190 to -101)	-28	-96,206	-187	+41.4	+129.9
UGA	Uganda	+48.40	+29.42	-3,361 (-5,512 to -2,108)	-1,186	-480,338	-8,462	+69.4	+114.3
TOTAL	All 15	0.00	-5.77	-1,866 (-4,437 to -1,150)	-1,013	+1,447,387	-10,746		

Step 5. Efficiency: the marginal effect of US$1M of 2027 US MOU funding

Method: for each country, the Original MOU run minus a run with the country's 2027 MOU amount lowered by US$1M (central parameters, everything else unchanged). 'People kept' are the extra full-cost equivalents on each service in 2027; 'deaths averted' are the extra deaths over 2026-2030 (and 2031-2035) that the US$1M prevents. A +US$1M run checks linearity. The model's own US$1M cards (`health_model.per_million_card`, audit I-04: a uniform US$1M-a-year cut for five years to one disease across all donors and channels) are shown afterwards for comparison; they are per disease, per year of money, and not scaled by the MOU leverage, so they are not the same quantity.

5a. Ranking by deaths averted per US$1M of 2027 MOU money (nominal)

Rank	Country	Leverage (model $ per MOU $)	Modelled aid per US$1M, US$M	ART kept	TB kept	Bednets kept	Spraying kept	Malaria Rx kept	Vaccines kept	All six services kept	Deaths averted 2026-30 per US$1M	Further 2031-35	Deaths averted per US$1M modelled aid	Check: +US$1M side	HIV infections averted
1	Botswana	0.90	0.90	1,366	0	6	1	0	0	1,373	138.3	64.3	153.6	138.3	1099.7
2	Nigeria	0.53	0.53	1,075	46	21,388	794	984	49	24,337	93.9	28.9	178.5	93.9	167.4
3	Mozambique	0.56	0.56	1,059	47	3,730	611	297	29	5,772	82.6	27.8	146.3	82.6	201.2
4	Eswatini	0.70	0.70	1,419	8	7	2	0	0	1,435	79.3	35.5	113.0	147.9	277.8
5	Malawi	0.65	0.65	1,379	2	7,845	618	807	5	10,657	77.0	30.7	119.0	77.0	180.5
6	Cameroon	0.57	0.57	1,254	5	20,361	304	669	14	22,607	75.5	35.3	132.6	75.5	190.0
7	Lesotho	0.64	0.64	1,299	1	9	2	1	0	1,312	73.7	47.2	115.9	73.7	523.9
8	Rwanda	0.46	0.46	597	0	12,443	1,741	416	5	15,202	66.5	22.1	144.9	66.5	150.3
9	Ethiopia	0.39	0.39	381	44	7,743	1,366	1,184	33	10,749	61.0	11.9	154.7	61.0	49.7
10	Kenya	0.65	0.65	914	16	8,784	314	572	15	10,615	57.4	17.5	88.8	57.4	87.0
11	Uganda	0.61	0.61	801	6	6,941	1,346	827	3	9,924	53.6	20.1	88.3	53.6	148.3
12	Sierra Leone	0.32	0.32	272	0	23,056	1,809	1,257	130	26,523	39.3	7.1	123.1	39.2	47.5
13	Madagascar	0.23	0.23	0	0	18,379	1,630	1,972	4	21,986	22.4	0.0	98.0	22.4	0.2
14	Burundi	0.34	0.34	324	0	4,873	573	220	1	5,992	19.5	7.3	57.6	19.5	46.6
15	Liberia	0.33	0.33	111	0	13,462	452	506	24	14,555	17.3	3.4	52.6	17.3	16.1

5b. Drivers

Leverage = the country's 2021-23 US direct-channel aid in the model / `mou_us_ref`: how many modelled dollars one nominal MOU dollar moves. Disease mix = share of the country's US direct-channel aid by disease (the cut is uniform across the US direct rows, so the money lost follows this mix). Service money per US$1M = the year's extra loss landing on each service line after the model splits unspecified and systems money

(`health_model.run_country` effective()). Unit costs are the central values the model uses (2023 US$). Deaths per unit = extra deaths 2026-2030 per extra person losing the service in 2027.

Disease mix of US direct aid and service money per US$1M of MOU money (US$ thousands by line; the rest is non-service money: programme management, systems, OVC and testing money beyond the kappa share)

Country	Leverage	US mix HIV	US mix TB	US mix Malaria	US mix Immunization	ART	HIV testing & care	HIV prevention	PMTCT	OVC	TB DS	Bednets	IRS	Malaria Rx	Vaccines
Botswana	0.90	99%	0%	0%	0%	254	161	229	3	17	2	0	0	0	0
Nigeria	0.53	73%	7%	19%	1%	203	120	20	5	15	13	55	8	16	5
Mozambique	0.56	82%	11%	7%	1%	229	109	70	10	7	27	13	6	5	3
Eswatini	0.70	95%	5%	0%	0%	261	116	99	6	36	21	0	0	0	0
Malawi	0.65	89%	2%	9%	0%	297	51	94	5	9	4	20	6	13	2
Cameroon	0.57	78%	2%	20%	0%	278	82	22	3	19	3	53	3	11	1
Lesotho	0.64	99%	0%	0%	0%	199	185	95	12	28	1	0	0	0	0
Rwanda	0.46	78%	0%	21%	0%	170	34	58	5	14	0	32	17	15	1
Ethiopia	0.39	67%	10%	23%	1%	103	76	17	1	15	20	35	14	20	3
Kenya	0.65	87%	4%	8%	0%	249	165	62	12	16	5	23	3	9	2
Uganda	0.61	84%	6%	10%	0%	243	106	75	14	17	3	18	13	14	0
Sierra Leone	0.32	37%	1%	57%	4%	48	27	18	0	0	0	60	18	21	13
Madagascar	0.23	3%	2%	95%	0%	0	0	1	0	0	0	48	16	32	0
Burundi	0.34	68%	0%	32%	0%	78	72	9	7	11	0	13	6	4	0
Liberia	0.33	39%	4%	57%	1%	29	47	5	1	0	0	35	5	8	2

Unit costs (US$ per person-year, course, net, person sprayed, case treated, child) and health effect per person losing the service

Country	ART $/py	TB DS $/course	Bednet $	IRS $/person	Malaria Rx $/case	Vaccines $/child	Deaths 2026-30 per ART py lost	per TB course lost	per 1,000 bednets lost	per 1,000 malaria cases untreated	per 1,000 children unvaccinated	HIV transmission /unsuppressed py	ART coverage	TB-HIV share	Malaria deaths per 1,000 cases	Bednet use	U5MR	Vaccine deaths averted per 1,000 children
Botswana	208	1,232	2.20	8.50	14.00	87	0.1001	0.491	0.000	n/a	n/a	0.308	100%	44%	0.92	50%	33	33.3
Nigeria	208	246	2.20	8.50	14.00	87	0.0510	0.454	0.495	4.76	40.75	0.122	78%	5%	2.68	28%	116	59.0
Mozambique	227	500	2.20	8.50	14.00	87	0.0507	0.468	0.327	2.97	22.10	0.130	84%	23%	1.79	43%	59	30.4
Eswatini	191	246	2.20	8.50	14.00	87	0.0511	0.527	0.000	0.00	n/a	0.137	95%	55%	0.00	50%	45	34.1
Malawi	199	500	2.20	8.50	14.00	87	0.0504	0.498	0.267	2.36	20.37	0.098	89%	46%	1.21	59%	49	27.8
Cameroon	216	246	2.20	8.50	14.00	87	0.0512	0.485	0.281	1.93	26.37	0.118	73%	16%	1.29	48%	65	39.4
Lesotho	191	246	2.20	8.50	14.00	87	0.0535	0.535	0.000	0.00	26.73	0.251	96%	58%	n/a	50%	60	36.3
Rwanda	208	500	2.20	8.50	14.00	87	0.0995	0.427	0.133	9.97	13.05	0.126	94%	12%	6.10	42%	38	18.9
Ethiopia	314	397	2.20	8.50	14.00	87	0.0975	0.453	0.138	1.57	23.25	0.073	82%	6%	1.34	9%	44	34.9
Kenya	308	273	2.20	8.50	14.00	87	0.0498	0.337	0.064	5.64	13.99	0.070	91%	21%	3.03	44%	39	20.9
Uganda	314	500	2.20	8.50	14.00	87	0.0506	0.462	0.251	3.51	18.73	0.125	85%	33%	1.35	68%	49	26.9
Sierra Leone	191	500	2.20	8.50	14.00	87	0.0508	0.475	0.637	5.29	28.79	0.132	84%	14%	3.04	83%	90	51.3
Madagascar	400	500	2.20	8.50	14.00	87	0.0527	0.469	0.472	6.65	16.45	0.120	24%	4%	5.15	42%	62	24.8
Burundi	299	500	2.20	8.50	14.00	87	0.0505	0.467	0.296	2.40	22.82	0.110	89%	6%	1.31	40%	47	29.6
Liberia	400	500	2.20	8.50	14.00	87	0.0514	0.480	0.498	7.30	34.08	0.111	59%	10%	3.48	61%	86	64.8

Caps and nonlinearity:

- Lines capped at the people covered (Original MOU run): none.
- Lines where donor money buys more than the donor-funded pool, so units scale with the share of money lost (`donor_exceeds_cost`): Botswana tb_ds; Botswana tb_dr; Botswana mal_cm; Botswana imm; Nigeria hiv_pmtct; Mozambique tb_dr; Mozambique mal_itn; Eswatini tb_ds; Eswatini tb_dr; Eswatini mal_itn; Eswatini mal_cm; Eswatini imm; Malawi tb_ds; Malawi tb_dr; Malawi imm; Cameroon hiv_pmtct; Cameroon tb_ds; Cameroon tb_dr; Lesotho hiv_pmtct; Lesotho tb_ds; Lesotho tb_dr; Rwanda hiv_art; Rwanda hiv_pmtct; Rwanda tb_ds; Rwanda tb_dr; Rwanda mal_cm; Ethiopia hiv_pmtct; Ethiopia tb_dr; Ethiopia mal_itn; Kenya hiv_pmtct; Uganda tb_dr; Uganda imm; Madagascar hiv_pmtct; Burundi hiv_pmtct; Burundi tb_ds; Burundi tb_dr; Liberia hiv_pmtct.
- Countries where the -US$1M and +US$1M marginals differ by more than 2% (nonlinear: malaria Lives Saved Tool, ART cohort ageing, TB spread): Eswatini (79.27 vs 147.88).

5c. The model's own US$1M cards (audit I-04), for comparison

Deaths 2026-2030 from a US$1M-a-year loss for five years to one disease (all donors and channels), and the main service's people losing it per year (HIV: ART; TB: TB treatment; malaria: bednets; immunization: routine vaccines).

Country	HIV deaths per US$1M/yr	TB deaths per US$1M/yr	Malaria deaths per US$1M/yr	Immunization deaths per US$1M/yr	HIV people per US$1M/yr	TB people per US$1M/yr	Malaria people per US$1M/yr	Immunization people per US$1M/yr
Botswana	273.3	219.5	2.1	0.0	1,392	96	79,592	0
Nigeria	425.7	2703.7	765.0	1873.6	2,144	1,303	161,507	9,770
Mozambique	384.0	1741.9	393.1	1016.1	1,788	805	107,413	9,770
Eswatini	364.5	409.3	0.0	436.9	1,827	168	39,625	3,997
Malawi	414.9	693.8	411.5	307.0	2,083	303	136,367	3,181
Cameroon	437.8	1529.2	342.1	1204.8	2,067	689	143,618	9,770
Lesotho	385.6	1004.3	0.0	1233.9	1,899	410	65,519	9,770
Rwanda	261.0	465.7	435.4	595.4	1,304	231	129,543	9,770
Ethiopia	222.4	2335.0	203.5	1063.4	1,099	1,118	78,353	9,770
Kenya	290.2	2559.9	580.8	644.2	1,364	1,204	150,197	9,770
Uganda	292.9	1274.3	497.1	469.5	1,285	586	125,121	5,320
Sierra Leone	336.3	1559.0	687.3	1312.0	1,543	707	138,408	9,770
Madagascar	131.5	1440.6	709.2	749.5	573	661	101,837	9,770
Burundi	186.1	1660.6	309.5	1049.2	912	765	96,365	9,770
Liberia	159.8	1464.0	554.5	1555.4	723	647	100,811	9,770

Step 5 (continued). Explanation

Where the money goes (table 4e)

All three ceilings take money from the same three countries and give most of it to Mozambique:

Scenario	Countries giving (nominal US$M)	Countries receiving (nominal US$M)
Low Ceiling	Nigeria -21.8, Kenya -19.4, Cameroon -3.3 (total -44.5)	Mozambique +43.0, Burundi +1.5
Base Ceiling	Nigeria -43.7, Kenya -38.8, Cameroon -6.6 (total -89.1)	Mozambique +77.3, Uganda +3.9, Sierra Leone +3.6, Lesotho +2.8, Burundi +1.5
Higher Ceiling	Nigeria -65.5, Kenya -58.2, Cameroon -9.9 (total -133.6)	Mozambique +77.3, Uganda +48.4, Sierra Leone +3.6, Lesotho +2.8, Burundi +1.5

Botswana, Eswatini, Ethiopia, Liberia, Madagascar, Malawi and Rwanda are unchanged in every scenario, so their results do not move.

What drives the differences (numbers from tables 4b, 4e, 5a and 5b)

1. **Deaths fall because the money moves to countries where a dollar averts more deaths than in Kenya.** Per US$1M of nominal MOU money, Mozambique averts 89-95 deaths over 2026-2030 where it arrives (table 4e), Lesotho 74 and Uganda 54-69, against 51-57 lost in Kenya and 56-76 in Cameroon where it leaves. Nigeria is close to Mozambique (87-94), so the Nigeria-to-Mozambique part of the move is roughly even; the gain comes mainly from Kenya's money. Kenya's money averts fewer deaths per dollar because its ART cost is high (US$308 per patient-year against US$227 in Mozambique) and about a sixth of each MOU dollar goes to HIV testing and care, which has no modelled deaths (US$165K per US$1M, table 5b). Net: Low Ceiling 3,858 deaths averted where money arrives against 3,412 caused where it leaves (-446); Base 7,913 against 6,778 (-1,134); Higher 11,067 against 9,200 (-1,866).
2. **People losing a service rise because Nigeria's money is bednet money.** One US$1M of Nigeria's MOU money keeps 21,388 people on bednets (US$55K of it lands on bednets at US$2.20 each), against 3,730 in Mozambique (table 5a). So moving money out of Nigeria removes many cheap bednet places and moving it into Mozambique adds few. Nigeria alone adds 531,373 / 1,062,747 / 1,594,120 people losing a service in 2027 (Low / Base / Higher), more than the whole net increase (+554,556 / +1,077,462 / +1,447,387); almost all of it is bednets (+536,777 / +1,033,463 / +1,428,931, table 4c). Each bednet place lost costs about 0.5 deaths per 1,000 in Nigeria, against 0.05 deaths per ART patient-year lost, so the people count rises while deaths fall. The services gained are mainly ART, PMTCT and HIV prevention: HIV deaths fall by 432 / 1,402 / 2,845 and HIV infections by 3,240 / 7,174 / 10,746, while malaria deaths rise by 338 / 622 / 849.
3. **Some modelled aid is lost in the conversion, and some money in Mozambique buys nothing.** The model converts each MOU amount into a cut fraction of the country's 2021-23 US direct-channel aid, so one MOU dollar is worth 0.65 modelled dollars in Kenya, 0.57 in Cameroon and 0.53 in Nigeria, but 0.56 in Mozambique, 0.61 in Uganda, 0.64 in Lesotho, 0.34 in Burundi and 0.32 in Sierra Leone (table 5a). In addition, the Base and Higher Ceilings give Mozambique US$481.0M, US$8.1M above its 2021-25 reference (US$472.9M): its cut is already 0% there, so that US$8.1M averts nothing in the model (Step 2). Together these mean the 15 countries lose US$1.1M / US$7.0M / US$5.8M more modelled aid in 2027 than under the Original MOU, even though the nominal total is the same. Mozambique's surplus is about US$4.6M of the US$7.0M in the Base Ceiling.
4. **Timing.** The gains build up over time. In 2027 itself the Higher Ceiling has 234 more deaths than the Original MOU (malaria and TB deaths from the money leaving Nigeria, Kenya and Cameroon happen in the same year), but from 2028 it has about 700 fewer each year, because deaths among people who stop ART rise in the years after stopping (table 4d). All three ceilings also reduce deaths in 2031-2035 (-201 / -485 / -1,013).
5. **Ranges.** The paired ranges of the change in deaths are below zero in all three ceilings (Low -1,062 to -318, Base -2,730 to -843, Higher -4,437 to -1,150), so the direction holds across the parameter draws. No line is capped at the people covered, and only Eswatini (unchanged here) is nonlinear in the marginal check.

In plain language

1. The new allocations take US MOU money from Kenya, Nigeria and Cameroon and give most of it to Mozambique, and in the Higher Ceiling a large share to Uganda.
2. In the model, Mozambique and Uganda use that money mostly for HIV treatment and prevention at lower cost per patient than Kenya, so the move averts more deaths than it causes: 446 fewer deaths over 2026-2030 under the Low Ceiling, 1,134 under the Base Ceiling and 1,866 under the Higher Ceiling (0.2-1.0% of the 189,629 under the Original MOU).
3. At the same time, many more people lose a service in 2027 (+0.55M to +1.45M), almost all of them bednets in Nigeria. Bednets are cheap per person, so losing them affects many people, but each lost net costs fewer lives than a person losing HIV treatment.
4. Malaria deaths go up a little and HIV deaths and infections go down; the net effect on deaths is a gain.
5. Two things to fix in the allocations: the Base and Higher Ceilings give Mozambique US$8.1M more than its reference, which the model treats as buying nothing; and Kenya's money averts the fewest deaths per dollar among the three countries giving money, so taking more from Kenya and less from Nigeria would avert more deaths, while taking more from Nigeria cuts more bednets.
6. This reverses the earlier allocations (allocations_2027.csv, RESULTS_2027.md), which sent money to malaria-heavy countries (Madagascar, Burundi, Liberia, Sierra Leone) where one MOU dollar is worth only 0.23-0.34 modelled dollars, and added 707 to 2,327 deaths.